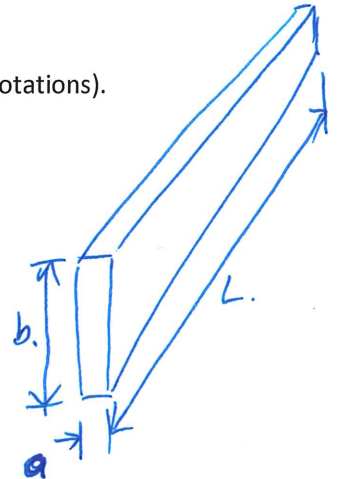
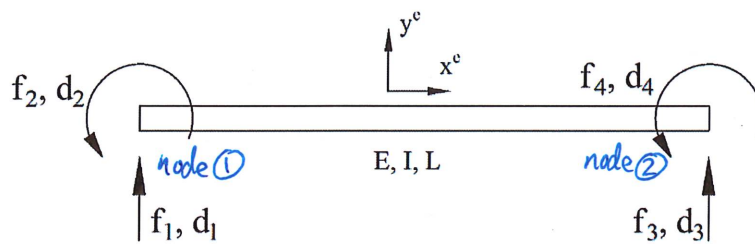


BEAM ELEMENTS

So far, we have only considered pin-jointed bar elements that only carry axial loads. We will now consider beam elements, which carry shear loads and moments. *only, in the absence of any axial loads.*

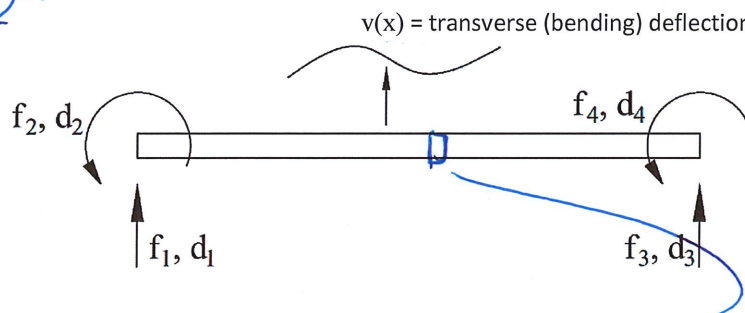
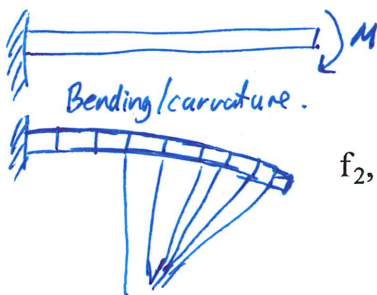
(1) Beam elements:

- Are defined by length, L , elastic modulus, E , and the second moment of area of the cross-section for bending, I .
- Carry moment and shear.
- Have **no axial component**, as they are considered **rigid axially**.
- Have 4 local DOF per element (2 transverse (shear) deflections and 2 rotations).

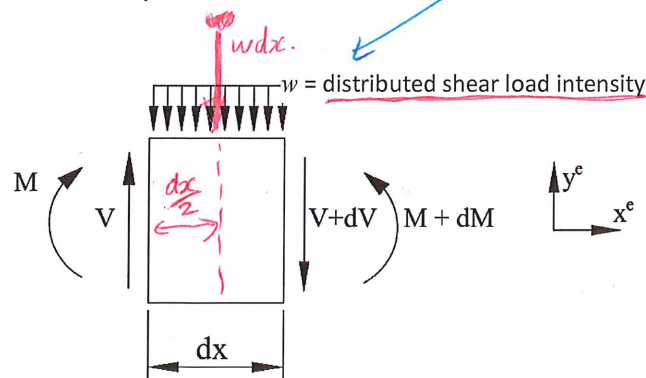
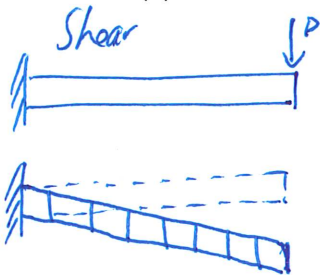


(2) Beam elements are formulated assuming the beam is "slender", $L \gg (a, b)$, where a and b are the cross-sectional dimensions.

generally $L > 5-10 \times$ the dimension b (or larger) allows us to use the Euler-Bernoulli beam bending theory.



(4) Consider a small finite element in equilibrium:



If the beam is slender, then bending deflections dominate and shear deflections are negligible.

(a) Force equilibrium in y^e axis:

$$\sum F_y = V - (V + dV) - w dx = 0$$

$$w = \frac{dV}{dx}$$

the transverse distributed load is equal to the rate of change of internal shear force.

(b) Bending moment equilibrium:

about the left edge of the element.

$$\sum M_z = -M - (w dx) \frac{dx}{2} - (V + dV) dx + (M + dM) = 0$$

$$-\frac{1}{2} w dx^2 - V dx - dV dx + dM = 0$$

Assuming the length of the selected segment, dx , is infinitesimally small, the following assumptions can be made:

$$dx^2 \rightarrow 0$$

$$dV dx \rightarrow 0$$

Assuming that 2nd order terms (small number squared) terms can be neglected.

Which leads to:

The value of the shear force is equal the rate of change of bending moment.

$$\frac{dM}{dx} = V$$

⇒ Linked to the small deflection assumption.

(c) From ENME 202 we know $M = EI \frac{d^2v}{dx^2}$ (the moment-curvature relationship).

(d) Hence, shear force can be expressed:

$$V = \frac{dM}{dx} = \frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right)$$

(e) Similarly, the uniformly distributed load intensity can be approximated:

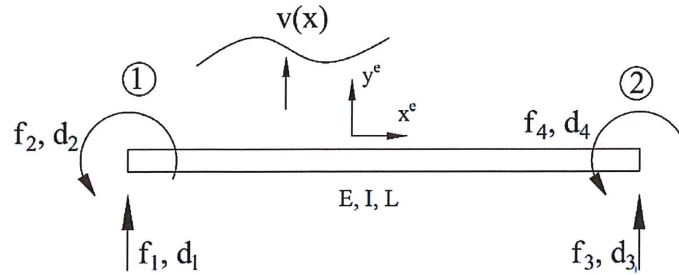
$$w = \frac{dV}{dx} = \frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right)$$

(5) As we did before for axial bar elements, we will assume external uniformly distributed load is equal to 0 (for now):

$$w(x) = 0$$

$$\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) = 0$$

(6) Hence, given E, I, f_1, f_2, f_3, f_4 over an element, find the deformation function, $v(x)$, such that:



$EI \frac{d^2v}{dx^2} = \text{moment}$
evaluated at $x=0$
 $\Rightarrow f_2$

$$\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) = 0$$

$EI \frac{d^3v}{dx^3} = \text{shear force.}$

$$EI \frac{d^3v}{dx^3} \Big|_{x=0} = f_1$$

Boundary Conditions (BCs) at node 1

$$-EI \frac{d^2v}{dx^2} \Big|_{x=0} = f_2$$

Shear force at node ②

$$-EI \frac{d^3v}{dx^3} \Big|_{x=L} = f_3$$

Boundary Conditions (BCs) at node 2

$$EI \frac{d^2v}{dx^2} \Big|_{x=L} = f_4$$

Bending moment applied at node ②.

(7) Using the principle of virtual displacement (PVD) and multiplying by a virtual displacement:

$$\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) = 0 \rightarrow \delta v \left(\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) \right) = 0$$

$$\int_0^L \underbrace{\delta v}_{g(x)} \left(\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) \right) dx = 0$$

(8) Integration by parts twice, the first time being:

Integrate by parts once

$$\int_0^L \delta v \left(\frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right) \right) dx = 0$$

Recall the formula for integration by parts

$$\int g(x) h'(x) dx = g(x) h(x) - \int h(x) g'(x) dx$$

In this instance, we will define:

$$g(x) = \delta v$$

$$g'(x) = \frac{d}{dx} \delta v$$

$$h(x) = \left(\frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right) \right)$$

$$h'(x) = \frac{d^2}{dx^2} \left(EI \frac{d^2v}{dx^2} \right)$$

$$\delta v \left(\frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right) \right) \Big|_0^L - \int_0^L \frac{d}{dx} \delta v \frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right) dx = 0$$

$$\left(\delta v(L) EI \frac{d^3v}{dx^3} \Big|_{x=L} - \delta v(0) EI \frac{d^3v}{dx^3} \Big|_{x=0} \right) - \int_0^L \frac{d}{dx} \delta v \frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right) dx = 0$$

$$-\delta v(0) f_1 - \delta v(L) f_3 - \int_0^L \frac{d}{dx} \delta v \frac{d}{dx} \left(EI \frac{d^2v}{dx^2} \right) dx = 0$$

$$= -f_3$$

Integrate by parts a second time:

$$-\delta v(0)f_1 - \delta v(L)f_3 - \int_0^L \frac{d}{dx} \delta v \frac{d}{dx} \left(EI \frac{d^2 v}{dx^2} \right) dx = 0$$

$$-\delta v(0)f_1 - \delta v(L)f_3 - \left[\frac{d}{dx} \delta v EI \frac{d^2 v}{dx^2} \right]_0^L + \int_0^L \frac{d^2}{dx^2} \delta v \left(EI \frac{d^2 v}{dx^2} \right) dx = 0$$

Recall the formula for integration by parts

$$\int g(x) h'(x) dx = g(x) h(x) - \int h(x) g'(x) dx$$

This time, we will define:

$$g(x) = \frac{d}{dx} \delta v$$

$$g'(x) = \frac{d^2}{dx^2} \delta v$$

$$h(x) = EI \frac{d^2 v}{dx^2}$$

$$h'(x) = \frac{d}{dx} \left(EI \frac{d^2 v}{dx^2} \right)$$

Moment at node ②, f_4 Moment at node ①, f_2 .

$$-\delta v(0)f_1 - \delta v(L)f_3 - \left[\left(\frac{d}{dx} \delta v EI \frac{d^2 v}{dx^2} \right)_{x=L} - \left(\frac{d}{dx} \delta v EI \frac{d^2 v}{dx^2} \right)_{x=0} \right] + \int_0^L \frac{d^2}{dx^2} \delta v \left(EI \frac{d^2 v}{dx^2} \right) dx = 0$$

$$-\delta v(0)f_1 - \delta v(L)f_3 - [\delta v'(L)(f_4 + \delta v'(0)f_2)] + \int_0^L \delta v''(EI v'') dx = 0$$

$$-\delta v(0)f_1 - \delta v(L)f_3 - \delta v'(L)f_4 - \delta v'(0)f_2 + \int_0^L \delta v'' EI v'' dx = 0$$

The following “weak form” of internal virtual work of the beam is obtained:

$$\int_0^L \delta v'' EI v'' dx = \delta v(0)f_1 + \delta v'(0)f_2 + \delta v(L)f_3 + \delta v'(L)f_4$$

Which can also be written as:

$$\int_0^L \delta v'' EI v'' dx = (\delta v(0) \quad \delta v'(0) \quad \delta v(L) \quad \delta v'(L)) \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix}$$

these were our $\psi(x)$ for bars.

(9) Now, assume that deflection at any point along the beam can be calculated using a set of shape functions, N :

$$v(x) = N d \quad d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$

(10) Assume that:

$$N = [N_1(x) \quad N_2(x) \quad N_3(x) \quad N_4(x)]$$

- 4 individual shape function equations.

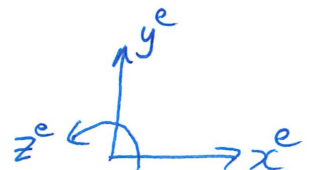
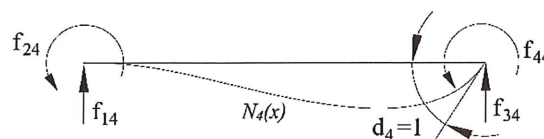
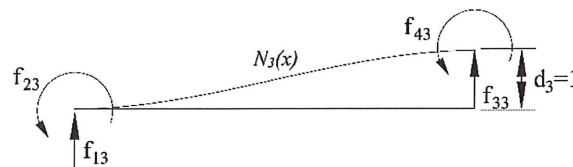
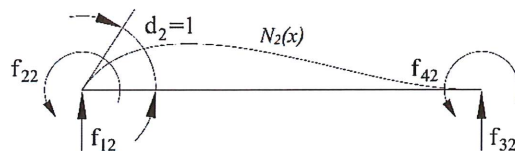
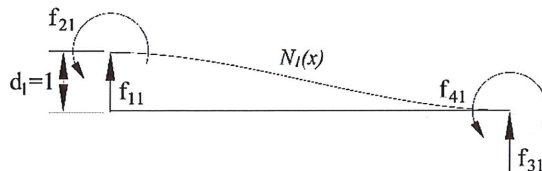
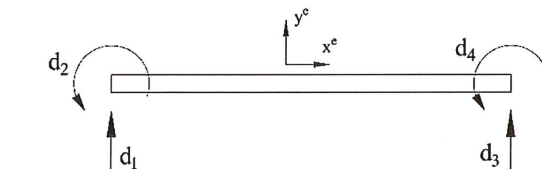
Where N can be expressed as a set of Hermitian polynomials:

$$N_i(x) = Ax^3 + Bx^2 + Cx + D$$

A generic 3rd order polynomial with 4 constants as yet undefined.

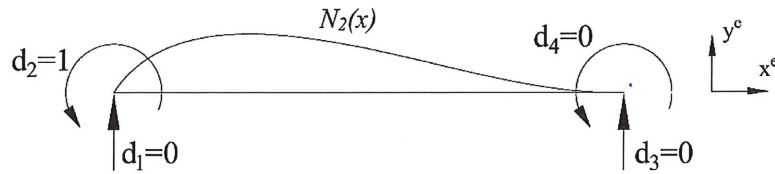
Where A, B, C, D are found from boundary conditions for each displaced shape below.

These four constants will be different for each shape.



Systematically work through and set each deflection component to 1.0 while keeping all others at zero.

(11) Shape function representing $d_2 = 1$ displacement shape can be derived as follows:



Begin with the general 3rd order polynomial: $N_i(x) = Ax^3 + Bx^2 + Cx + D$

Then apply the following boundary conditions:

Our four
boundary
conditions.

$N_2(x=0) = d_1 = 0$	— slope at $x=0$.
$\left. \frac{dN_2}{dx} \right _{x=0} = d_2 = 1$	
$N_2(x=L) = d_3 = 0$	
$\left. \frac{dN_2}{dx} \right _{x=L} = d_4 = 0$	

Using the boundary conditions:

BC1: $N_2(0) = A(0)^3 + B(0)^2 + C(0) + D = 0$
 $\downarrow$
 $D = 0$

BC2: $N'_2(0) = 3A(0)^2 + 2B(0) + C = 1$
 $\downarrow$
 $C = 1$

BC3: $N_2(L) = A(L)^3 + B(L)^2 + C(L) + D = 0$
 $\downarrow$
 $AL^2 + BL + 1 = 0$

BC4: $N'_2(L) = 3A(L)^2 + 2B(L) + C = 0$
 $\downarrow$
 $3AL^2 + 2BL + 1 = 0$

Constants A and B can be found using equations for boundary conditions BC3 and BC4, which yields:

$$A = \frac{1}{L^2} \quad B = -\frac{2}{L}$$

The following shape function is obtained:

$$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$$

$N_2(x)$ is the specific version of our generic 3rd order polynomial that captures the shape shown at the top of the page.

(12) Applying a similar approach for the other deflected shapes yields $N_1(x)$, $N_2(x)$, $N_3(x)$, $N_4(x)$:

$$\begin{aligned} N_1(x) &= 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3} \\ N_2(x) &= \frac{x^3}{L^2} - \frac{2x^2}{L} + x \\ N_3(x) &= \frac{3x^2}{L^2} - \frac{2x^3}{L^3} \\ N_4(x) &= \frac{x^3}{L^2} - \frac{x^2}{L} \end{aligned}$$

These are the four equations which model the four deflected shapes shown on page 67.

(13) Hence, the transverse deflection along the length of the beam from multiple deflection components can be given by:

The vector of the four shape functions above.
/ Solved nodal deflections.

$$v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$

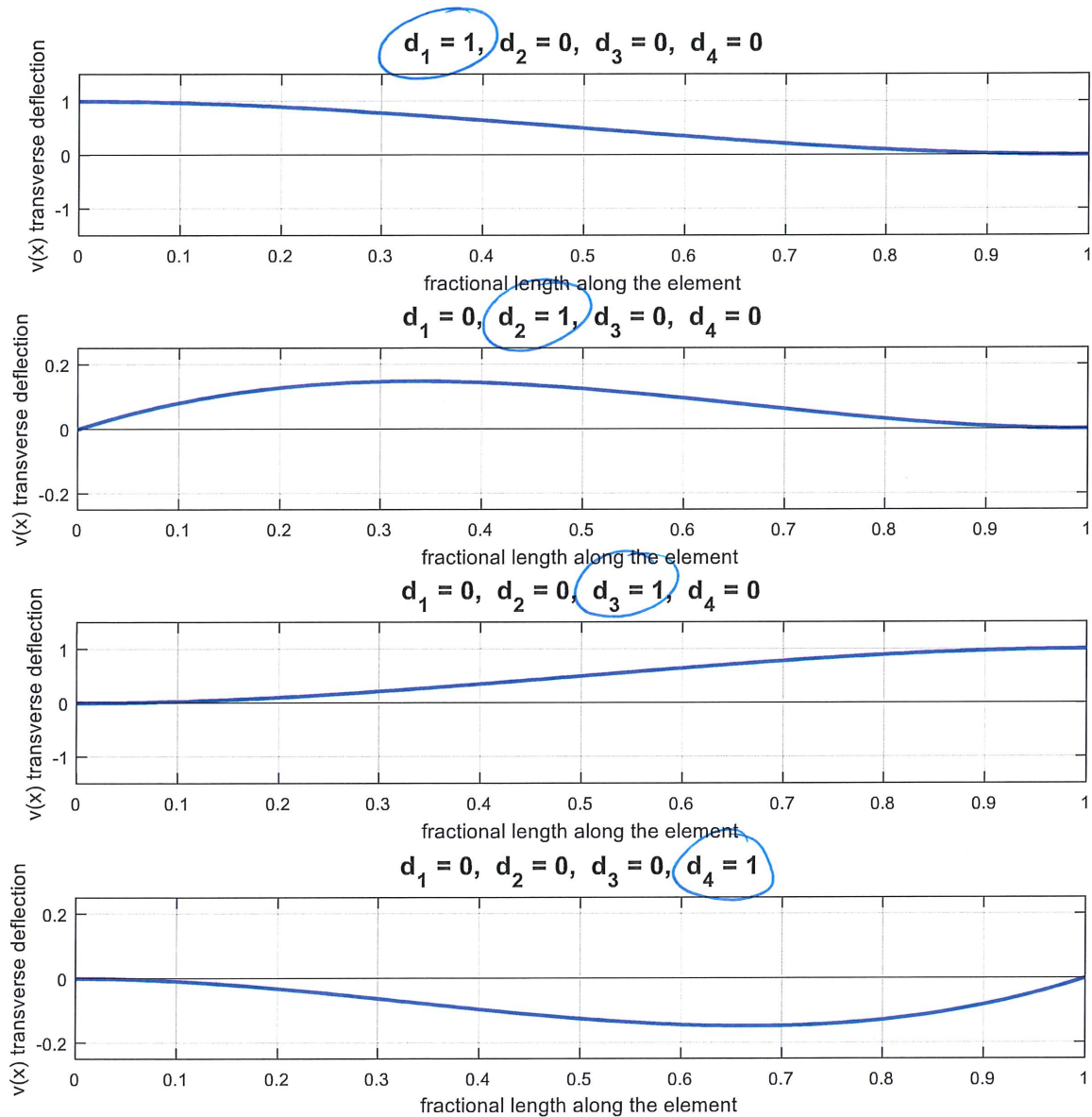
which yields:

$$v(x) = N_1(x)d_1 + N_2(x)d_2 + N_3(x)d_3 + N_4(x)d_4$$

(14) Shape functions are derived based on the following assumptions:

- "Small" motions
- Bernoulli-Euler beam theory, so aspect ratio is $>5-10$, meaning not applicable to squat, short beam elements.
- The transverse deflection profile, $v(x)$ can only be weighted combinations of d_1, d_2, d_3, d_4 using $N(x)$ shape functions.

Let's plot the four shape functions (set each of the four displacements to a value of 1.0, with all others zero, just like we used in the derivation of the shape functions)



$$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$$

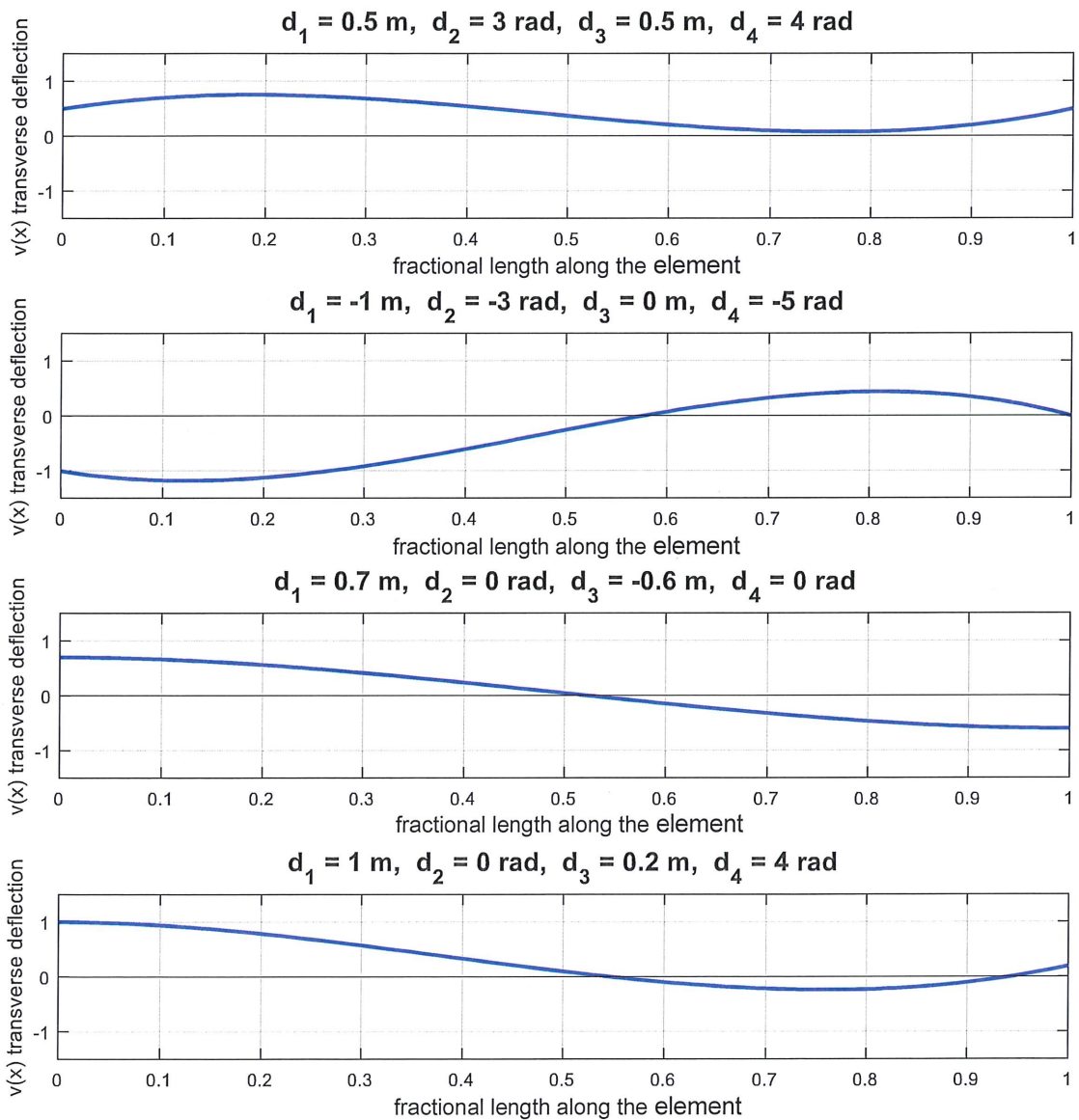
$$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$$

$$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$$

$$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$$

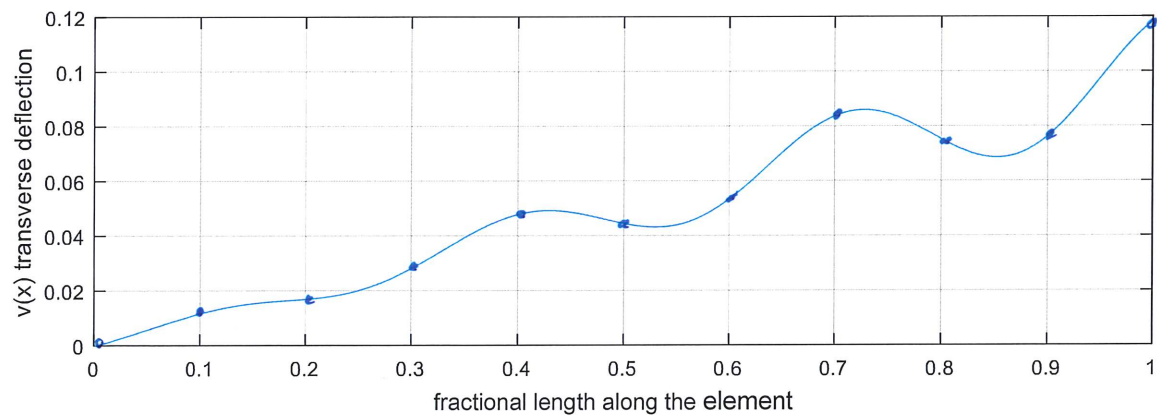
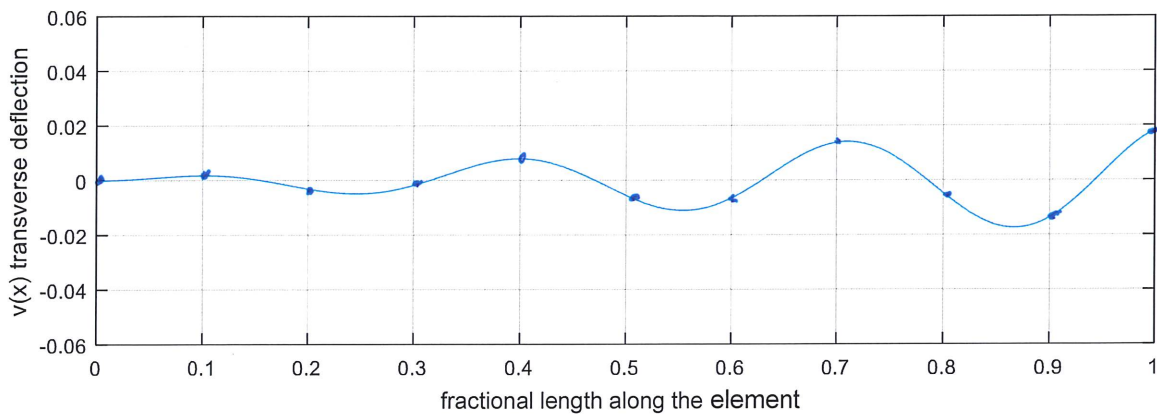
Below is some example nodal deflection values and the corresponding deflected shapes. The deflections are not small (so this violates our assumptions) but the deflections are exaggerated to display some of the allowable deflected shapes.

$$v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$



Do you think it is possible for the beam element that we have derived to capture the displaced profile below?

If you are modelling a structure where this sort of deflected profile is expected, what should you do?
What must you be careful of?



$\Rightarrow$ One element cannot capture this shape,
but multiple elements can.

One Simple Element.

2 nodes, 4 DOF per element.



A more complex element.

3 nodes per element
6 DOF per element.



These shape functions represent the internal reaction mechanisms that occur within the beam and enable us to relate loads at the ends of the element (the nodes) to the corresponding deflections.

We can now use these shape functions to define a stiffness matrix for this beam element, using a closely similar method to what we did for the bar element.

(15) Repeat and rewrite Equation (8)

$$\int_0^L \delta v'' EI v'' dx = \delta d^e T f^e$$

where:

essentially just $v(x) = v d^e$ but in an incremental.

$$\delta v = N \delta d^e \quad \text{and} \quad \delta v'' = N''(x) \delta d^e$$

substituting:

$$\int_0^L \left(\delta d^e T (N''(x))^T EI (N''(x)) d^e \right) dx = \delta d^e T f^e$$

$$\delta d^e T \left(\int_0^L (N''(x))^T EI (N''(x)) dx \right) d^e = \delta d^e T f^e$$

$$K^e = \int_0^L \left((N''(x))^T EI N''(x) \right) dx$$

where:

$$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$$

$$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$$

$$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$$

$$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$$

so that:

$$N_1''(x) = -\frac{6}{L^2} + \frac{12x}{L^3}$$

$$N_2''(x) = \frac{6x}{L^2} - \frac{4}{L}$$

$$N_3''(x) = \frac{6}{L^2} - \frac{12x}{L^3}$$

$$N_4''(x) = \frac{6x}{L^2} - \frac{2}{L}$$

We can apply the shape function associated with each deflected shape to generate a corresponding stiffness term, and in doing so we will generate the stiffness matrix for this element.

*look at the form of this equation.
 $(\dots) d^e = f^e$
 all of this stuff within the brackets represents stiffness*

(16) For example, ij^{th} component of K^e matrix can be obtained:

the stiffness term that sits in the i^{th} row and the j^{th} column of K^e .

We are assuming E and I are constant along the element.

$$K_{ij}^e = \int_0^L N_i''(x)^T EI N_j''(x) dx$$

For example,

$$\begin{aligned} K_{22}^e &= EI \int_0^L \left(\frac{6x}{L^2} - \frac{4}{L} \right) \left(\frac{6x}{L^2} - \frac{4}{L} \right) dx \\ &= EI \int_0^L \left(\frac{36x^2}{L^4} - \frac{48x}{L^3} + \frac{16}{L^2} \right) dx \\ &= EI \left[\frac{36x^3}{3L^4} - \frac{48x^2}{2L^3} + \frac{16x}{L^2} \right]_0^L \\ &= EI \left[\frac{36L^3}{3L^4} - \frac{48L^2}{2L^3} + \frac{16L}{L^2} \right] \\ &= EI \left[\frac{12}{L} - \frac{24}{L} + \frac{16}{L} \right] \\ &= \frac{4EI}{L} = \frac{EI}{3} (4L^2) \end{aligned}$$

the transpose of the 2nd derivative of the i^{th} shape function (w.r.t. x).

$$N_i''(x) = N_2(x) \quad N_j''(x) = N_2(x)$$

If we were to reverse the two bracketed terms, then we would get the same answer.

$$K_{ij} = K_{ji}$$

$\Rightarrow$ The stiffness matrix must be symmetrical.

(17) If we repeat the procedure above for all $i = 1 \dots 4$ and $j = 1 \dots 4$ possible motions, we will obtain beam element stiffness matrix in local (element co-ordinates), K^e :

$$K^e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

The stiffness matrix in element coordinates for a beam.

Note: $K_{ji} = K_{ij}$ meaning stiffness matrix is always symmetric.

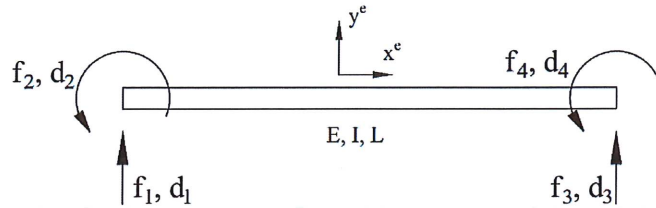
$$f^e = K^e d^e$$

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$

(18) Notes on beam element stiffness matrix:

Numbering sequence
for a beam element:

local y^e at node ① $\Rightarrow d_1$
local z^e at node ① $\Rightarrow d_2$
local y^e at node ② $\Rightarrow d_3$
local z^e at node ② $\Rightarrow d_4$



$$f^e = K^e d^e \rightarrow \text{in element coordinate system}$$

$$K^e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

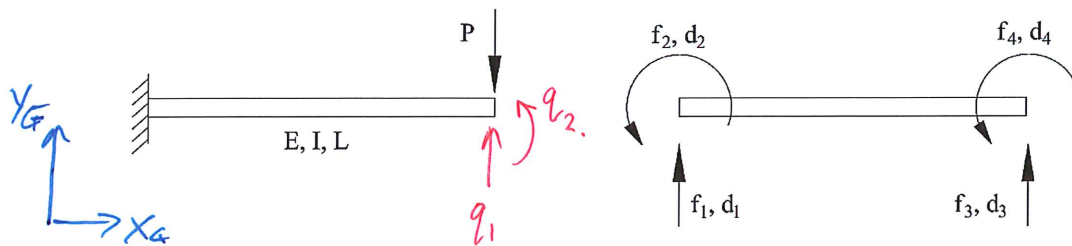
- The order of terms within this matrix is based upon the degree of freedom numbering in the diagram above. If we had numbered these degrees of freedom differently, the stiffness terms within this matrix would be in a different order. Therefore, to be able to use this stiffness matrix, we must follow the same numbering sequence that was used during the derivation.
- Term EI/L^3 represents the flexural rigidity of the beam, which in some texts is noted as:

$$\bar{\alpha} = \frac{EI}{L^3}$$

- Beam element stiffness matrix, K^e , is symmetric. This means force at one end has same effect at the other end, no matter which end is loaded.
- Stiffness matrix is derived assuming E and I are constant over beam length. If E or I varies the integral in Equation (15) must be for $E(x)$ and/or $I(x)$ resulting in a different stiffness matrix, K^e .
- Is valid for long, slender beams whose response/deformation is predominantly flexural with aspect ratio $> 5-10$.
- Accounts for small motions as stiffness matrix is derived ignoring dx^2 term in (4).
- Assumes beam is axially rigid, which implies $EA/L \gg EI/L^3$ which is typically true when the beam is long and slender so that L is relatively large.
- K^e is not "full-rank" as boundary conditions have not yet been applied and the element is therefore not constrained (just like with bar elements).
 - For a 4x4 matrix, "full-rank" means a rank of 4.
 - This "rank" parameter is an indication of the number of linearly independent rows (or columns) in the matrix, which corresponds to the number of unique equations. A rank less than the size of a matrix (not full rank) means that it cannot be solved for an independent/unique solution.

$\gg$ np. findg. matrix-rank(K^e)

Beam Element Example 1. Simple cantilever beam to prove the concept



(1) Global forces can be calculated:

$$f^e = K^e d^e$$

$$A' = \begin{bmatrix} D'_1 & D'_2 & D'_3 & D'_4 \\ q_1 & 0 & 0 & 1 \\ q_2 & 0 & 0 & 0 \end{bmatrix}$$

$$K_G^e = A^e \hat{K}^e A^{eT}$$

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$

However, cantilever beam shown above is rigidly fixed on the left side, meaning $d_1 = d_2 = 0$ and f_1, f_2 will be reaction loads. As a result, we can ignore first 2 rows and columns of the K^e matrix (ie: ignoring the stiffness terms which relate to the zero displacement d_1 and the zero rotation d_2 at the fixed support), leading to:

$$\begin{pmatrix} f_3 \\ f_4 \end{pmatrix} = \begin{pmatrix} -P \\ 0 \end{pmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & -6L \\ -6L & 4L^2 \end{bmatrix} \begin{pmatrix} d_3 \\ d_4 \end{pmatrix}$$

Displacements, d , can be found by re-arranging this equation:

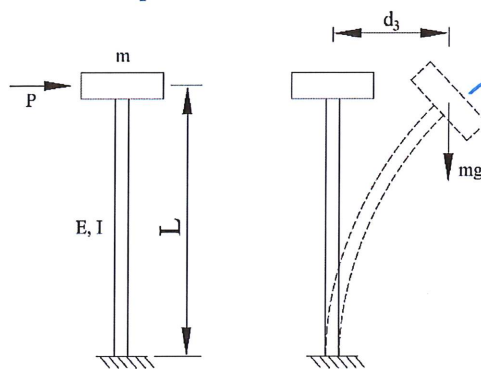
$$\begin{pmatrix} d_3 \\ d_4 \end{pmatrix} = \frac{L^3}{EI} \frac{1}{48L^2 - 36L^2} \begin{bmatrix} 4L^2 & 6L \\ 6L & 12 \end{bmatrix} \begin{pmatrix} -P \\ 0 \end{pmatrix} = \frac{L^3}{EI} \frac{1}{12L^2} \begin{pmatrix} -4L^2P \\ -6LP \end{pmatrix} = \frac{L}{12EI} \begin{pmatrix} -4L^2P \\ -6LP \end{pmatrix}$$

$$\begin{pmatrix} d_3 \\ d_4 \end{pmatrix} = \begin{pmatrix} \frac{-PL^3}{3EI} \\ \frac{-PL^2}{2EI} \end{pmatrix}$$

vertical tip deflection = $\frac{-PL^3}{3EI}$
tip rotation = $\frac{-PL^2}{2EI}$

Which matches analytical solutions in books for tip deflection d_3 and rotation d_4 of a tip-loaded cantilever that you would have got in ENME202.

Beam Element Example 2. Vertical cantilever beam with lumped mass on top



the mass displaces outside the footing and creates an overturning moment.

If we consider this (very simplified) 2 DOF building (or bridge pier) model.

What have we neglected?

Answer: Gravity!

Ignoring axial effects at rest/in the undeformed shape, the gravity force mg has no impact. But deflected, it adds a further base moment:

$$M = mg \times d_3$$

which will increase deflection d_3 and thus, we would have to iterate to get a solution.

If instead, we simply said:

$$\begin{pmatrix} d_3 \\ d_4 \end{pmatrix} = \frac{L}{12EI} \begin{bmatrix} 4L^2 & 6L \\ 6L & 12 \end{bmatrix} \begin{pmatrix} P \\ mg \frac{PL^3}{3EI} \end{pmatrix}$$

$$d_3 = \frac{PL^3}{3EI} \left(1 + \frac{mgL^2}{2EI} \right)$$

The added term $\frac{mgL^2}{2EI}$ can amount to 20-30% more displacement!

Consider if:

$$\begin{aligned} E &= 200 \text{ GPa} \\ I &= 2 \times 10^{-5} \text{ m}^4 \end{aligned}$$

$$\begin{aligned} L &= 6 \text{ m} \\ P &= 10 \text{ kN} \end{aligned}$$

$$mg = 40 \text{ kN}$$

Horizontal displacement due to horizontal force alone:

$$d_3 = \frac{PL^3}{3EI} = \frac{(10 \times 10^3) (6^3)}{3(200 \times 10^9)(2 \times 10^{-5})} = 0.18 \text{ m}$$

Horizontal displacement due to horizontal force AND gravity force:

$$d_3 = \left(1 + \frac{mgL^2}{2EI} \right) \frac{PL^3}{3EI} = \left(1 + \frac{(40 \times 10^3)(6^2)}{2(200 \times 10^9)(2 \times 10^{-5})} \right) \frac{(10 \times 10^3) (6^3)}{3(200 \times 10^9)(2 \times 10^{-5})} = 0.212 \text{ m}$$

- Hence, addition of gravity forces results in 18% increase in horizontal displacement.
- In earthquake terms a 10 kN lateral load in a 40kN building is $\sim 0.25g$ peak ground acceleration (PGA), which is a relatively large earthquake.
- A 0.5g earthquake would then have increased this effect.
- Again, it is a non-linear problem because $d_3 = f(d_3)$ and an iterative solution is required, but the effect can be much larger.